Weinstein tubular neighborhood for the contact form, with an explicit collar model

The Univalent Foundations Program

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Editorial note. The counted statement is the author's labelled lemma 'l:Weinstein' (source lines 715-738); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own displayed equation for the normalised primitive on the collar, which the statement itself invokes by label.

Editorial note. The retained counter values (section 2, theorem offset 9) make the counted lemma print as Lemma 2.10; this article carries no label-number dump, so the number was inferred by counting the theorem-class environments of section 2 and is reported with that caveat.

Editorial note. No retained line refers to a label, citation or figure outside the retained spans, so the package declares no external reference.

Editorial note. The retained preamble is a minimal editorial adapter supplying only the plain text abbreviations that the publisher class used to define, followed by the native author macro block of the pinned source.

Editorial scope. One original result of the pinned publisher article, counted once: the Weinstein tubular neighbourhood theorem for the contact form with its collar model (author label l:Weinstein).

Allowed standard prerequisites (editorial).

- Canonical Liouville form on the cotangent bundle. lambda_std is the standard Liouville one-form on T*Sigma, with d lambda_std the canonical symplectic form.. It is the target form whose pull-back is computed at source lines 754-761 and 802.
- Collar coordinates and the normalised primitive. On the collar A_C = [0,rho) x S^1 the primitive nu_0 has the displayed form with a constant a_C.. It is used to write f_0 explicitly at source lines 765-772.
- Moser's trick. Given a family of forms mu_t with mu_0 = mu_1 up to exactness one solves X_t interior dmu_t = -(d/dt)mu_t and integrates to an isotopy pulling mu_1 back to mu_0 up to an exact form.. It is the whole second half of the proof, source lines 777-808.

$$\nu_0|_{A_C} = \left(a_C - \tfrac{1}{2}r^2\right) ds, \tag{1}$$

Lemma 2.10 (Weinstein tubular neighborhood)**.** *There exists an open neighborhood $U \subset \Sigma \times \Sigma$ of the diagonal $\Delta_\Sigma = \{(z,z) \mid z \in \Sigma\}$, an open neighborhood $V \subset \mathrm{T}^*\Sigma$ of the 0-section $0_\Sigma \subset \mathrm{T}^*\Sigma$, and a smooth map $\psi : U \to V$ that restricts to a diffeomorphism $\psi : U \cap \mathrm{int}(\Sigma \times \Sigma) \to V \cap \mathrm{T}^*\mathrm{int}(\Sigma)$, and satisfies*

$$\psi(\Delta_\Sigma) = 0_\Sigma, \qquad \psi^*\lambda_{\mathrm{std}} = \nu_2 - \nu_1 + df,$$

where $f : \Sigma \times \Sigma \to \mathbb{R}$ is a smooth function such that $f|_{\Delta_\Sigma} \equiv 0$ and

$$df(z_1,z_2) = (\nu_1 - \nu_2)_{(z_1,z_2)}, \qquad f(z_1,z_2) = -\int_{[z_1,z_2]} \nu_0,$$

for every pair $(z_1,z_2) \in U \cap (\partial\Sigma \times \partial\Sigma)$. If $A_C \equiv [0,\rho) \times S^1$ is the collar neighborhood of a connected component C of $\partial\Sigma$ on which ν_0 has the form (1), the restriction $\psi|_{U\cap(A_C\times A_C)}$ has the form

$$\begin{aligned} \psi(r,s,R,S) = \big(R,\, s,\, R\,(S-s),\, \tfrac{1}{2}(r^2-R^2)\big), \\ \forall (r,s,R,S) \in U \cap (A_C \times A_C). \end{aligned} \tag{2}$$

Proof. We first provide the construction within the collar neighborhood $A = A_C$ of each connected component C of $\partial\Sigma$. We consider a small enough neighborhood $W \subset A \times A$ of the diagonal $\Delta_A = \{(z,z) \mid z \in A\}$ so that, for each $(r,s,R,S) \in W$, the points $s, S \in S^1$ are not antipodal. We define the map

$$\kappa_0 : W \to \underbrace{\mathrm{T}^*A}_{A\times\mathbb{R}^2}, \qquad \kappa_0(r,s,R,S) = \big(R,\, s,\, R\,(S-s),\, \tfrac{1}{2}(r^2-R^2)\big).$$

This map restricts to a diffeomorphism onto its image

$$\kappa_0 : W \cap \mathrm{int}(A \times A) \to \mathrm{int}(\mathrm{T}^*A),$$

and satisfies $\kappa_0(\Delta_A) = 0_A$ and

$$\begin{aligned} \kappa_0^*\lambda_{\mathrm{std}} &= R\,(S-s)\,dR + \tfrac{1}{2}(r^2-R^2)\,ds \\ &= -\tfrac{1}{2}\,R^2\,dS + \tfrac{1}{2}\,r^2\,ds + d\big(\tfrac{1}{2}(S-s)\,R^2\big) \\ &= -\nu_1 + \nu_2 + d\big((a_C - \tfrac{1}{2}R^2)(s-S)\big) \\ &= -\nu_1 + \nu_2 + df_0, \end{aligned}$$

where

$$f_0 : W \to \mathbb{R}, \qquad f_0(r,s,R,S) = (a_C - \tfrac{1}{2}R^2)(s-S).$$

Notice that $f_0|_{\Delta_A} \equiv 0$ and

$$\begin{aligned} f_0(0,s,0,S) &= a_C(s-S) = -\int_{[z_1,z_2]} \nu_0, \\ df_0(0,s,0,S) &= a_C\,ds - a_C\,dS = \nu_1 - \nu_2, \end{aligned}$$

where $z_1 = (0,s)$ and $z_2 = (0,S)$.

For some sufficiently small neighborhood $U \subset \Sigma \times \Sigma$ of the diagonal Δ_Σ, we choose an arbitrary smooth function $f_1 : U \to \mathbb{R}$ that coincides with f_0 on a neighborhood of $U \cap (\partial\Sigma \times \partial\Sigma)$ and satisfies $f_1|_{\Delta_\Sigma} \equiv 0$ and $df_1 = \nu_1 - \nu_2$ at all points of the diagonal Δ_Σ. Up to further shrinking the neighborhood

U, we also choose a smooth map $\kappa_1 : U \to \mathrm{T}^*\Sigma$ that coincides with κ_0 on a neighborhood of $W \cap (\partial\Sigma \times \partial\Sigma)$, restricts to a diffeomorphism onto its image $\kappa_1 : U \cap \mathrm{int}(\Sigma \times \Sigma) \to \mathrm{T}^*(\mathrm{int}(\Sigma))$, and such that $\kappa_1(\Delta_\Sigma) = 0_\Sigma$.

We now conclude the proof by means of a typical application of Moser's trick. We set

$$\mu_t := t\kappa_1^*(\lambda_{\mathrm{std}}) + (1-t)(\nu_2 - \nu_1),$$

and we look for an isotopy $\phi_t : U \to \Sigma \times \Sigma$, defined after possibly further shrinking the neighborhood U, such that $\phi_0 = \mathrm{id}$ and $\phi_t^*\mu_t - \mu_0$ is exact. We denote by X_t the time-dependent vector field generating ϕ_t and compute

$$\begin{aligned}\tfrac{d}{dt}\phi_t^*\mu_t &= \phi_t^*(L_{X_t}\mu_t + \tfrac{d}{dt}\mu_t) = \phi_t^*\big(X_t \lrcorner\, d\mu_t + d(\mu_t(X_t)) + \mu_1 - \mu_0\big)\\ &= \phi_t^*\big(X_t \lrcorner\, d\mu_t + \kappa_1^*\lambda_{\mathrm{std}} - \nu_2 + \nu_1 - df_1 + d(\mu_t(X_t) + f_1)\big).\end{aligned}$$

Notice that the symplectic forms $\kappa_1^* d\lambda_{\mathrm{std}}$ and $-d\nu_1 + d\nu_2$ define the same orientation, since they coincide on a neighborhood of $U \cap (\partial\Sigma \times \partial\Sigma)$. Therefore $d\mu_t$ is symplectic away from $\partial(\Sigma \times \Sigma)$ for every $t \in [0,1]$. We choose the vector field X_t so that

$$X_t \lrcorner\, d\mu_t + \kappa_1^*\lambda_{\mathrm{std}} - \nu_2 + \nu_1 - df_1 = 0.$$

Notice that X_t vanishes on the diagonal Δ_Σ and on a neighborhood of $U \cap (\partial\Sigma \times \partial\Sigma)$. Moreover

$$\tfrac{d}{dt}\phi_t^*\mu_t = \phi_t^* d(\mu_t(X_t) + f_1).$$

Up to shrinking the neighborhood U, we obtain a well defined isotopy $\phi_t : U \to \mathrm{T}^*\Sigma$ that coincides with κ_0 on a neighborhood of $U \cap (\partial\Sigma \times \partial\Sigma)$ and satisfies

$$\phi_1^*\kappa_1^*(\lambda_{\mathrm{std}}) = -\nu_1 + \nu_2 + df,$$

where

$$f(w) = \int_0^1 \big(\mu_t(X_t) + f_1\big) \circ \phi_t(w)\, dt.$$

The desired map is $\psi := \kappa_1 \circ \phi_1$. □